

Mathematical Symbols and Constants

Fall 2025

Symbol	Definition	Example
$\mathbb{N}$	Set of natural numbers	$\mathbb{N} = \{1, 2, 3, \dots\}$
$\mathbb{N}_0$	Natural numbers including zero	$\mathbb{N}_0 = \{0, 1, 2, 3, \dots\}$
$\mathbb{Z}$	Set of integers	$\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$
$\mathbb{Q}$	Set of rational numbers	$\frac{1}{2} \in \mathbb{Q}$
$\mathbb{R}$	Set of real numbers	$\sqrt{2} \in \mathbb{R}$
$\mathbb{C}$	Set of complex numbers	$1 + i \in \mathbb{C}$
$\in$	Element of a set	$x \in \mathbb{R}$
$\notin$	Not element of a set	$\pi \notin \mathbb{Q}$
$\subset, \subseteq$	Subset, subset or equal	$A \subseteq B$
$\supset, \supseteq$	Superset, superset or equal	$B \supseteq A$
$\cap$	Intersection of sets	$A \cap B$
$\cup$	Union of sets	$A \cup B$
$\emptyset$	Empty set	$A \cap \emptyset = \emptyset$
$\forall$	For all (universal quantifier)	$\forall x \in \mathbb{R}, x^2 \geq 0$
$\exists$	There exists (existential quantifier)	$\exists x \in \mathbb{R} : x^2 = 4$
$\Rightarrow$	Implies	$x > 1 \Rightarrow x^2 > 1$
$\Leftrightarrow$	If and only if (iff)	$x^2 = 1 \Leftrightarrow x = \pm 1$
$ A $	Cardinality of a set	If $A = \{1, 2, 3\}$, then $ A = 3$
$\mathbb{P}(A)$	Probability of event A	$\mathbb{P}(A) = 0.3$
$\mathbb{E}[X]$	Expectation of random variable X	$\mathbb{E}[X] = \sum x_i p_i$
$\text{Var}(X)$	Variance of X	$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$
σ	Standard deviation	$\sigma = \sqrt{\text{Var}(X)}$
Δx	Change or difference in x	$\Delta x = x_2 - x_1$
$\partial f / \partial x$	Partial derivative of f wrt x	$\partial(xy) / \partial x = y$
∇f	Gradient of f	$\nabla f = (\partial f / \partial x, \partial f / \partial y)$
$\int f(x) dx$	Integral of $f(x)$	$\int x dx = \frac{x^2}{2} + C$
$\sum_{i=1}^n a_i$	Summation notation	$\sum_{i=1}^3 i = 6$
$\prod_{i=1}^n a_i$	Product notation	$\prod_{i=1}^3 i = 6$
π	Archimedes' constant	$\pi \approx 3.1416$
e	Euler's number (base of $\ln$)	$e \approx 2.718$

Table 1: Mathematical Symbols, Definitions, and Examples

Symbol	Definition	Example
i	Imaginary unit	$i^2 = -1$
∞	Infinity	$\lim_{x \rightarrow \infty} \frac{1}{x} = 0$
$\approx$	Approximately equal to	$\pi \approx 3.14$
$\equiv$	Identically equal to	$\sin^2 x + \cos^2 x \equiv 1$
$\propto$	Proportional to	$y \propto x^2$
$\lfloor x \rfloor$	Floor (greatest integer $\leq x$)	$\lfloor 2.7 \rfloor = 2$
$\lceil x \rceil$	Ceiling (smallest integer $\geq x$)	$\lceil 2.7 \rceil = 3$
$\binom{n}{k}$	Binomial coefficient	$\binom{4}{2} = 6$

Table 2: Mathematical Symbols, Definitions, and Examples